

Final:

- check Brightspace

Thursday (April 16), 3:30-6:30 pm

Toldeo Health Education ctr 102

- covers lectures 1-23 (today is 23)
- open book (any documents, no electronics)
- questions (mostly similar to midterm format)
- exam starts at 3:30 and takes about 90 minutes
- bring some photo id

$$a_n + a_{n-1} - 4a_{n-2} - 4a_{n-3} = 0$$

$$a_0 = 8, a_1 = 6, a_2 = 26$$

$$\textcircled{1} \quad x^n + x^{n-1} - 4x^{n-2} - 4x^{n-3} = 0$$

$$\downarrow \div x^{n-3}$$

$$x^3 + x^2 - 4x - 4 = 0$$

$$\textcircled{2} \quad x^2(x+1) - 4(x+1) = 0$$

$$(x+1)(x^2 - 4) = 0$$

$$(x+1)(x-2)(x+2) = 0$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$x_1 = -1 \quad x_2 = 2 \quad x_3 = -2$$

difference of squares

$$a^2 - b^2 = (a-b)(a+b)$$

$$\textcircled{3} \quad a_n = \alpha(-1)^n + \beta(2)^n + \gamma(-2)^n$$

$$\textcircled{4} \quad a_0 = 8 \Rightarrow a_0 = \alpha + \beta + \gamma = 8$$

$$a_1 = 6 \Rightarrow a_1 = -\alpha + 2\beta - 2\gamma = 6$$

$$a_2 = 26 \Rightarrow a_2 = \alpha + 4\beta + 4\gamma = 26$$

$$3\beta - \gamma = 14$$

$$6\beta + 2\gamma = 32$$

$$\times 2 \rightarrow 6\beta - 2\gamma = 28$$

$$\rightarrow 6\beta + 2\gamma = 32$$

$$12\beta = 60 \Rightarrow \beta = 5$$

$$\gamma = 1$$

$$\alpha = 2$$

$$a_n = 2(-1)^n + 5(2)^n + 1(-2)^n$$

H.w. $a_n = 6a_{n-1} - 9a_{n-2}$ $a_0 = 1, a_1 = 2$

$\Rightarrow x_1 = 3, x_2 = 3$

$\Rightarrow a_n = \alpha (3)^n + \beta n(3)^n$

$\Rightarrow \alpha = 1, \beta = 1$

linear non-homogeneous recurrences :

$\alpha_0 a_n + \alpha_1 a_{n-1} + \alpha_2 a_{n-2} + \dots + \alpha_k a_{n-k} = f(n)$

non-homo.
↓

coefficients : constant numbers

general strategy :

$a_n = a_n^{(h)} + a_n^{(p)}$

↓
homogeneous solution
obtain as before

↓
particular solution
guess
(similar to $f(n)$)

	$f(n)$	$a_n^{(p)}$
constant	$\rightarrow c$	$A \leftarrow \text{constant}$
	n	$An + B$
	n^2	$An^2 + Bn + C$
	r^n	Ar^n

$$a_n = 3a_{n-1} + 2n \quad a_1 = 3$$

$$a_n - 3a_{n-1} = 2n \quad \leftarrow f(n)$$

$$\text{solution } a_n = a_n^{(h)} + a_n^{(p)}$$

$$\underline{a_n^{(h)}}$$

$$a_n - 3a_{n-1} = 0$$

$$x - 3 = 0 \quad \text{char. eq.}$$

$$x = 3 \quad \text{root}$$

$$\boxed{a_n^{(h)} = c(3)^n}$$

$$\underline{a_n^{(p)}}$$

$$a_n^{(p)} = An + B$$

$f(n)$ is degree-1 $\leftarrow$

to find A, B we plug $a_n^{(p)}$ to original recurrence

$$a_n - 3a_{n-1} = 2n$$

$$(An+B) - 3(A(n-1)+B) = 2n$$

$$\underline{An+B} - \underline{3An+3A-3B} - \underline{2n} = 0$$

$$\underline{n(-2A-2)} + \underline{(3A-2B)} = 0$$

$$\downarrow$$

$$-2A-2=0$$

$$-2A=2$$

$$\boxed{A=-1}$$

$$\downarrow$$

$$3A-2B=0$$

$$\downarrow$$

$$-3-2B=0 \rightarrow -2B=3 \rightarrow$$

$$\boxed{B = -\frac{3}{2}}$$

$$\boxed{a_n^{(p)} = -n + \frac{-3}{2}}$$

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$a_n = d(3)^n + (-n - \frac{3}{2})$$

$$a_1 = 3 \Rightarrow d(3)^1 + (-1 - \frac{3}{2}) = 3$$

$$3d - \frac{5}{2} = 3$$

$$3d = \frac{11}{2}$$

$$d = \frac{11}{6}$$

$$\Rightarrow a_n = \frac{11}{6}(3)^n - n - \frac{3}{2}$$

rec
↓

$$a_2 = 3a_1 + 2 \times 2 = 3 \times 3 + 2 \times 2 = 13$$

non-rec
↓

$$a_2 = \frac{11}{6}(3)^2 - 2 - \frac{3}{2} = \frac{99}{6} - 2 - \frac{3}{2}$$

$$= \frac{99 - 12 - 9}{6} = \frac{78}{6} = 13$$

$$H.W. \quad a_n = a_{n-1} + 7n, \quad a_0 = 4$$

$$a_n - 2a_{n-1} + a_{n-2} = \overbrace{2^n}^{f(n)} \quad a_0 = 1, a_1 = 2$$

answer $a_n = a_n^{(h)} + a_n^{(p)}$

$a_n^{(h)}$: $a_n - 2a_{n-1} + a_{n-2} = 0$

$$x^2 - 2x + 1 = 0$$

$$(x-1)(x-1) = 0 \quad \left\{ \begin{array}{l} x_1 = 1 \\ x_2 = 1 \end{array} \right.$$

$$\boxed{a_n^{(h)} = \alpha(1)^n + \beta n(1)^n}$$

$a_n^{(p)}$: $f(n) = 2^n \xrightarrow{\text{guess}} a_n^{(p)} = A 2^n$

↓ plug into original recurrence

$$a_n - 2a_{n-1} + a_{n-2} = 2^n$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$A2^n - 2(A2^{n-1}) + A2^{n-2} = 2^n$$

$$\cancel{A}2^n - \cancel{A}2^n + A2^{n-2} = 2^n$$

$$\boxed{A = 2^2 = 4}$$

$$\Rightarrow a_n^{(p)} = 42^n \Rightarrow \boxed{a_n^{(p)} = 2^{n+2}}$$

$$a_n = \alpha(1)^n + \beta n(1)^n + 2^{n+2}$$

$$a_0 = 1 \Rightarrow \alpha + 2^2 = 1 \Rightarrow \boxed{\alpha = -3}$$

$$a_1 = \alpha + \beta + 8 = 2 \Rightarrow (-3) + \beta + 8 = 2 \Rightarrow \boxed{\beta = -3}$$

$$\Rightarrow a_n = -3 - 3n + 2^{n+2}$$